Literature Status for the Rudelson–Vershynin L-infinity Entropy Exponent Question

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Status

This note records that a specific open entropy-exponent question in Rudelson–Vershynin, arXiv:math/0404192, is answered in later literature by Aiyer–Mansour–Moran–Shao–Waknine, arXiv:2605.13684.

No new theorem is claimed here. The purpose of this packet is to clear an open-problem signal from the queue and point to the exact later result.

Source Question

In Section 4 of arXiv:math/0404192, after Theorem 4.4, Rudelson and Vershynin prove the empirical L_∞ entropy estimate

$$D_\infty(\mathcal{F}, t) \leq Cv \log(n/vt)\, \log^\varepsilon(2n/v), \qquad v = \mathrm{vc}(\mathcal{F}, c\varepsilon t).$$

They compare this with the Alon et al. bound, which gave $D_\infty(\mathcal{F}, t) = O(\log^2 n)$ in the relevant finite-domain regime, and then state that it remains open whether the logarithmic exponent can be made 1.

The local source evidence is PDF page 21 of `source_paper.pdf`; in the parsed arXiv TeX source, this sentence occurs at `data/parsed/arxiv_sources/0404192/source.tex`, line 1459.

Later Answer

The later paper arXiv:2605.13684 explicitly identifies the same Rudelson–Vershynin question. Its abstract says that the paper obtains sharp empirical ℓ_∞ metric-entropy bounds and that these results resolve open questions by Alon et al. and Rudelson–Vershynin.

The theorem-level statement is Theorem 2. For a bounded real-valued function class $\mathcal{F} \subset [-R, R]^X$, let

$$\gamma^* = \inf\{\gamma > 0 : \mathrm{fat}_\gamma(\mathcal{F}) < \infty\}.$$

Then the empirical metric entropy $H(\mathcal{F}, \gamma, n)$ is $O(\log n)$ for $\gamma \in (2\gamma^*, R)$. The theorem also records the lower regimes, including a sometimes tight $O(\log^2 n)$ behavior below the critical scale.

Immediately after Theorem 2, the authors say that, in view of the $O(\log^{1+\varepsilon} n)$ bounds of Rudelson and Vershynin, their result resolves the question of whether the exponent can be improved to 1. Therefore the source question is already answered by later literature, and the supporting authors explicitly knew they were answering it.

Scope

This packet only clears the L_∞ empirical entropy exponent-one question following Theorem 4.4 of arXiv:math/0404192. It does not claim to settle other open-problem signals or neighboring geometric questions in the source paper.

The later answer is scale-sensitive: exponent-one growth holds above the threshold in Theorem 2, while a near-$\log^2 n$ lower bound remains in the lower regime. This sharp threshold behavior is part of the resolution.

References

1. M. Rudelson and R. Vershynin, *Combinatorics of random processes and sections of convex bodies*, arXiv:math/0404192; Annals of Mathematics 164 (2006), 603–648.

2. S. Aiyer, Y. Mansour, S. Moran, H. Shao, and T. Waknine, *Scale-Sensitive Shattering: Learnability and Evaluability at Optimal Scale*, arXiv:2605.13684.